Qudit-based quantum simulation of fermionic systems

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Model Hamiltonians described by fermionic operators are ubiquitous in physics and chemistry, but the quantum simulation of their properties is challenging on near-term quantum computers. Indeed, for any problem with more than two connected neighbors, standard mappings of fermions on qubits lead to many-body operators whose size increases with the size of the problem, quickly making the simulation unfeasible on noisy devices. Here we consider an alternative mapping based on qudits and the auxiliary fermion method, which leads to a significant reduction in the circuit depth and in the number of computational units to be controlled. These advantages increase for highly connected target Hamiltonians, such as lattices with long-range interactions or the molecular systems investigated in quantum chemistry. We numerically test our approach on qudits encoded in molecular spins, by performing the quantum simulation of a fermionic model on a small cluster.

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I. INTRODUCTION

Quantum computers are typically based on binary building blocks called qubits, which use two states of a physical system to encode quantum information. This is often realized in practice by restricting a higher dimensional Hilbert space to its lowest-energy levels, thus overlooking potentially useful resources. Therefore, it is natural to envisage a computational model based on higher-dimensional units known as qudits [1–5]. Several architectures such as photons [6,7], ion traps [8,9], superconducting systems [10–14], and molecular nanomagnets [15–18] have already been proposed. Recent achievements are the use of trapped ions as qudits with up to seven levels [9], and the first prototype quantum simulator based on molecular spin qudits [19].

An approach based on qudits offers several advantages. Qudits have the potential to store and process more information per unit, resulting in simpler experimental setups with fewer physical objects. Furthermore, a computation based on qudits can lead to simpler and shallower circuits by reducing the number of two-body gates and by increasing the number of operations that can be done in parallel. Considering near term devices, these features could greatly facilitate the implementation of basic algorithmic protocols [5,9,20], for instance quantum error correction (QEC) [21–27] and digital quantum simulations [28–31].

Here we focus on the quantum simulation of fermionic systems, which is crucial in many fields ranging from condensed matter physics [32,33] to high-energy physics [34] and quantum chemistry [35]. Indeed, simulating the Fermi-Hubbard model and its many extensions could be the key to unravel phenomena which are still not understood (e.g., those related to high-temperature superconductivity) and to design novel and more performant molecules and materials. However, the quantum simulation of fermionic models is highly inefficient using the standard Jordan-Wigner (JW) or Braviy-Kitaev (BK) fermion to spin mapping [36,37]. Indeed, in more than one dimension or for arbitrary connectivity between the orbitals in the target model, this requires the simulation of multiqubit terms in which the number of Pauli operators increases with the size of the problem. This translates into very deep circuits which are not affordable on near-term devices.

In order to simplify the mapping and thus the implementation cost, researchers recently proposed using qudits with a specific dimension corresponding to the natural algebra of fermion models [31,38,39]. In the same vein, we have considered the auxiliary fermion method (AFM) [40–42], originally formulated for qubits, and shown how it can be efficiently implemented on qudits. The AFM maps fermionic operators to qubit operators with a fixed size, independent of the system's overall size. The price is an increase in the number of qubits compared to standard approaches. In our qudit AFM, these multiple qubits are merged into a single qudit spanning the same Hilbert space. This encoding reduces the number of units to be experimentally controlled compared to the qubit version, and requires only one- and two-qudit gates, thus reducing the overall circuit depth and its duration. We demonstrate the effectiveness of this approach by numerical simulations on a qudit architecture consisting of molecular nanomagnets (MNMs) [29,43,44]. These clusters contain a small number of coupled transition metal spins, resulting in

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a multilevel structure with high degree of chemical control, combined with the possibility to coherently manipulate these levels via microwave or radio-frequency pulses [22,24,44,45]. Different nanomagnets could be coupled either by placing them within superconducting resonators and exploiting the exchange of photons [43], or by using supramolecular structures where the strength of interactions fits a suitable hierarchy [44,46–49]. The latter strategy is explored here, with an interposed spin 1/2 acting as a dynamical switch of the interaction between molecular qudits. We investigate schemes to implement two-qudit gates in such a supramolecule, which are then exploited for simulating a proof-of-principle experiment of a fermion model on a cluster. We obtain good results with realistic parameters, showing that the proposed qudit encoding applied to molecular spins could be a promising route for the simulation of fermionic models.

The paper is organized as follows: in Sec. II we give an overview of the main fermion to qubit mappings. We then focus on the AFM and introduce our qudit encoding. In Sec. III we present a prototype architecture based on MNMs and illustrate different schemes to control them and to implement one- and two-qudit operations. We then numerically test the proposed scheme for the quantum simulation of a spinless fermion model on the elementary unit of a square lattice (Sec. IV), finding good results for reasonable values of the qudit coherence times. We finally draw the conclusions in Sec. V.

II. QUANTUM SIMULATION OF FERMIONIC SYSTEMS

In this work, we focus on the Fermi-Hubbard Hamiltonian, which can be written as

$$\begin{aligned}\mathcal{H} &= \mathcal{H}_t + \mathcal{H}_U \\ &= -\sum_{i,j}\sum_{\sigma} t_{ij} a^\dagger_{i,\sigma} a_{j,\sigma} + \sum_i U_i a^\dagger_{i,\uparrow} a_{i,\uparrow} a^\dagger_{i,\downarrow} a_{i,\downarrow}, \end{aligned} \quad (1)$$

where $a^\dagger_{i,\sigma}$ and $a_{i,\sigma}$ are fermionic creation and annihilation operators on site i with spin $\sigma = \uparrow, \downarrow$, satisfying $\{a_{i,\sigma}, a^\dagger_{j,\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. The first term describes hopping between sites in the lattice (parameterized by the hopping integrals t_{ij}) and the second represents the Coulomb interaction between electrons localiszed on the same orbital. Extensions with more orbitals per site (e.g., the periodic Anderson model) or with Coulomb two-body terms involving different sites are also of interest. The time evolution operator is represented by the unitary matrix $e^{-i\mathcal{H}\tau}$, and a quantum simulation corresponds to implementing this unitary on the hardware. This approach involves two steps. First, the quantum system has to be mapped onto the hardware, i.e., the algebra of the fermionic operators must be rewritten in terms of qubit or qudit operators. This step is crucial as a well-designed mapping can greatly simplify the simulation. Then, the unitary operator resulting from this mapping is decomposed into a sequence of single- and two-body gates that can be directly implemented on the hardware. To simplify this step, the evolution operator $e^{-i\mathcal{H}\tau}$ is commonly approximated as the product of evolution operators containing local terms (see Appendix D). The following subsection provides an overview of the most common fermion to qubit mappings.

A. Qubit mappings

Jordan-Wigner (JW) [36,50] and Bravyi-Kitaev (BK) [51–56] are two standard techniques for mapping fermions to qubits. These methods share two important aspects: the number of qubits coincides with the number of fermionic orbitals, and local fermion operators are mapped onto nonlocal qubit operators. JW encodes the occupation numbers of fermions into the qubit states. Orbitals must be numbered according to a chosen ordering. Fermionic operators for the (spin-)orbital p are mapped as

$$a_p = \bigotimes_{k=0}^{p-1} Z_k \otimes A_p, \quad (2)$$

where $A_p = (X_p - iY_p)/2$ is the qubit lowering operator and X_p, Y_p, and Z_p are the Pauli operators. Hopping terms are rewritten as follows:

$$a^\dagger_p a_q = A^\dagger_p \otimes \left(\bigotimes_{k=p+1}^{q-1} Z_k \right) \otimes A_q. \quad (3)$$

The Z chain $\bigotimes_{k=p+1}^{q-1} Z_k$ is introduced to correctly reproduce the canonical anticommutation relations. This chain complicates the simulation when $p \neq q \pm 1$, i.e., p and q are not adjacent in the chosen ordering. For example, by adopting the standard snakelike numbering of orbitals, the number of terms in this chain (also known as Pauli weight) scales as $O(N)$ for a two-dimensional (2D) square lattice and as $O(N^2)$ in a 3D cubic lattice, where N is the linear size of a lattice (total spins N^2 for square, N^3 for cubic). The digital quantum simulation of the time evolution operator of the Pauli chain requires long sequences of two-body gates, which are very demanding for current noisy devices (see Appendix A) [37,57].

A complementary approach to JW uses the parity basis [58]. In this mapping, qubit i stores the parity information, i.e., the sum (modulo 2) of the occupation of all orbitals with index less than or equal to i. However, information on the occupation of electronic orbitals is delocalized. This approach does not solve the problem of nonlocal operations, but it shifts it from counting parity to counting occupation, resulting in the same scaling as JW.

The Bravyi-Kitaev mapping can be described as a hybrid between the JW (also known as occupation basis) and the parity basis. Indeed, some qubits are used to encode the occupation information of certain fermionic orbitals, while the others are used to encode the parity information relative to specific subsets of orbitals. The advantage of this approach is that a better scaling can be obtained, with operators size growing as $O(\log_2(N))$ in 2D and as $O(\log_2(N^2))$ in 3D. For a complete description of this method we refer the reader to Refs. [52,54].

The JW and Bravyi-Kitaev methods have been extensively studied, and several optimized variants have been proposed [57,59]. Both these mappings use a single qubit for each fermionic orbital. Relaxing this condition, the AFM maps fermionic operators to limited-length qubit operators whose Pauli weight does not depend on the lattice size, but only on the number of connections of each site. This reduces the weight of Pauli strings to be simulated, resulting in shallower circuits. Moreover, by reducing the number of qubits involved

TABLE I. Requirements and costs to simulate the Hubbard model for $S = 1/2$ fermions, Eq. (1), on an $N \times N$ square lattice with nearest (N.N.) or next-to-nearest neighbor (N.N.N.) hopping interactions exploiting Jordan-Wigner, Bravyi-Kitaev, and qubit and qudit auxiliary fermion method mappings. Two types of hardware connectivity are considered: all-to-all (All) and square grid (Grid). The top panel shows the number of physical units, i.e., the number of qubits or qudits, required. The middle panel reports the largest operator weight, i.e., the maximum number of operators involved in a hopping term. The bottom panel shows the depth a Trotter step in terms of two-body gates.

Number of hardware physical units				
	JW	BK	Qubit AFM	Qudit AFM
N.N.	$2N^2$	$2N^2$	$4N^2$	$2N^2$
N.N.N.	$2N^2$	$2N^2$	$8N^2$	$2N^2$

Largest operator weight				
	JW	BK	Qubit AFM	Qudit AFM
N.N.	$N+1$	$\propto \log(N)$	4	2
N.N.N.	$N+2$	$\propto \log(N)$	8	2

Depth of a Trotter step						
	JW	BK	Qubit AFM		Qudit AFM	
	All	All	All	Grid	All	Grid
N.N.	$\propto N^2$	$\propto N\log(N)$	30	42	5	17
N.N.N.	$\propto N^2$	$\propto N\log(N)$	94	118	9	33

in each operator, more terms can be implemented simultaneously, leading to reduction of circuit depth in practical settings. The AFM is substantially based on the JW transformation, which is applied to a modified Hamiltonian where additional fermionic orbitals are introduced to encode the parity of the interactions, eliminating the need of Z chains. For each Hamiltonian term of the original model that JW would map to a nonlocal qubit operator, i.e., all terms involving distant orbitals in the linear indexing, auxiliary orbitals and associated auxiliary operators M_{aux} are introduced. For example, a hopping term $a_p^\dagger a_q + a_q^\dagger a_p$ between nonconsecutive sites p and q is replaced by $a_p^\dagger M_{\text{aux}}(pq) a_q + a_q^\dagger M_{\text{aux}}^\dagger(pq) a_p$. The same approach can be applied to two-body (four-operators) or any higher-order coupling terms. These auxiliary operators take the general form $M_{aux}(p,q) = i b_{p'} c_{q'}$, where

$$b_{p'} = \left(e^{-i\theta} a_{p'} + e^{i\theta} a_{p'}^\dagger\right),$$
$$c_{q'} = \left(e^{-i\phi} a_{q'} + e^{i\phi} a_{q'}^\dagger\right). \tag{4}$$

M_{aux} operators only act on auxiliary orbitals (labeled by primed letters; see Appendix B).

The drawback of this strategy is the significant increase of required qubits. For instance, for a 2D-lattice with nearest-neighbor (or next-to-nearest-neighbor) interactions, two (four) times the number of units are needed. In Table I we report the necessary resources and circuit complexity in terms of two-body gates for Hamiltonian (1) (particles with spin $S = 1/2$) on a square 2D lattice of size N (total number of sites N^2) in the case of nearest-neighbor (N.N.) or next-to-nearest-neighbors (N.N.N.) hopping interactions. Let us start by comparing the JW, BK, and AFM methods on qubits (first three columns).

The top panel shows the number of required physical units, i.e.. the number of qubits. As explained above, the qubit AFM method introduces additional qubits which, for the models analyzed, leads to a doubling and quadrupling of the original number of hardware units. In the second panel we report the largest operator weight, i.e., the maximum number of qubit operators appearing in a term of $\mathcal{H}_t$. We clearly see the contrast between the AFM, that uses operators of limited size, and the JW, that introduces operators with unbounded size. The bottom panel shows the depth of a single Trotter step in terms of two-body gates. For the JW and BK mappings, we have assumed all-to-all connectivity between the qubits, although in practical implementations the actual connectivity is often limited due to technical constraints. As a result, many SWAP gates would be required to enable the required two-qubit interactions, drastically increasing the complexity of the circuit. The scaling of these procedures can be optimized by using nesting methods as reported in [56], although without eliminating the dependence on the size of the system.

In the case of the AFM, we consider the gates decomposition presented in Appendix A and two assumptions for the hardware connectivity: all-to-all (All) and square grid (Grid). The additional SWAP gates are counted as three two-body gates, corresponding to the standard CNOT decomposition for qubits. Since the depth of the circuit is independent of the size of the system, the larger the system is, the greater the advantage is over JW. However, the crossing point is for relatively small lattices $N \sim 7$.

Hereafter we investigate how a qudit mapping can overcome the growth in the number of objects and improve the performance of the AFM. In fact, a qudit encoding naturally avoids the growth in number of computational units and further reduces the depth of the circuits, allowing for many operations to be performed simultaneously.

B. Qudit auxiliary fermion method

The idea behind the auxiliary fermion method is to store parity information inside the extra degrees of freedom given by the auxiliary qubits (see Appendix B). The disadvantage of this approach is that, while drastically reducing the problematic Pauli chains, it increases the number of hardware units with respect to the JW or BW mappings. To solve this issue, here we propose to exploit qudits. Starting from the standard qubit AFM, we map the product space of each qubit together with its auxiliaries onto a single qudit $|q\rangle \otimes |q'\rangle \cdots = |Q\rangle$. In this way, the occupation and parity information relative to a site is stored into a single object. The dimension of the qudit $|Q\rangle$ depends on the number of auxiliary sites and thus on the connectivity of the system, but not on the size of the system. The most common and interesting lattice models involve interactions between nearest or next-nearest neighbors. Therefore, small- to medium-size qudits are sufficient for simulating two- and three-dimensional lattices (see Table I). For example, in the case of a 2D rectangular lattice with nearest-neighbor interactions, four-level qudits are enough. 16-level qudits are sufficient to extend to next-nearest interactions.

Looking at the last column of Table I one can explicitly see the advantages of this approach. First of all, the one-to-one relationship between fermion orbitals and hardware units is preserved. Second, the largest operator weight, or in other words, the number of operators involved in a single term, is further reduced compared to the qubit AFM approach, leading to additional circuit compression (in terms of two-body gates).

The evolution operators related to a hopping term in the Hamiltonian can be rewritten in terms of two-qudit generalized XY_{ij}^{lk} gates:

$$\begin{aligned}\mathrm{XY}_{ij}^{lk}(\theta) &= \cos(\theta/2)(|i,j\rangle\langle i,j| + |l,k\rangle\langle l,k|)\\ &+ i\sin(\theta/2)(|l,k\rangle\langle i,j| + |i,j\rangle\langle l,k|)\\ &+ \sum_{|r,t\rangle\neq|i,j\rangle,|l,k\rangle} |r,t\rangle\langle r,t|,\end{aligned} \tag{5}$$

where $|i,j\rangle, |l,k\rangle$ are two-qudit states. Conversely, the evolution induced by $e^{-i\mathcal{H}_U\tau}$ is represented by a diagonal matrix and it is decomposed into a single generalized CPHASE gate,

$$\mathrm{CPHASE}(\bar{\varphi}) = \sum_{i,j=1}^{d} e^{-i\varphi_{i,j}}|i,j\rangle\langle i,j|, \tag{6}$$

whose effect is to give a different relative phase to each pair of levels in the product space of the qudit. In the following paragraphs we consider a possible specific implementation of this qudit mapping on a MNM-based architecture, and we discuss the system requirements and the different possible strategies to implement the needed gates.

III. HARDWARE STRUCTURE

Molecular nanomagnets have attracted a growing interest in the last decade as potential qudit-based platforms for quantum technologies. Indeed, MNMs are often characterized by an addressable multilevel structure which can be tuned by bottom-up chemical engineering [47,49]. Moreover, the state of MNMs can be manipulated by microwave or radio-frequency pulses in times much shorter than the typical decoherence time T_2 [22,24,60]. Hence, the lowest energy levels of a single molecule can naturally encode a qudit [16,22,26,29,44,61].

We start by considering the simplest molecular qudit, consisting of a single spin S complex, which provides up to $2S+1$ levels. This can be described by a spin Hamiltonian of the form

$$H_Q = H_Z + H_{\mathrm{ZFS}} = \mu_B g_z B_0 S_z + D S_z^2, \tag{7}$$

where the first term is the Zeeman interaction with the external static magnetic field B_0 along z, while H_{ZFS} represents the axial zero field spitting, which is fundamental to guarantee the addressability of transitions as it makes gaps nonequivalent.

To build a scalable architecture, the ability to implement switchable two-body entangling gates is required. One possible strategy is to interpose an $S = 1/2$ unit, S_w, between two qudits Q_1 and Q_2. Weak S_w-Q_1 and S_w-Q_2 interactions can be exploited to mimic a switchable coupling between Q_1 and Q_2 [47,62,63] (see Fig. 1). The Hamiltonian of the whole system

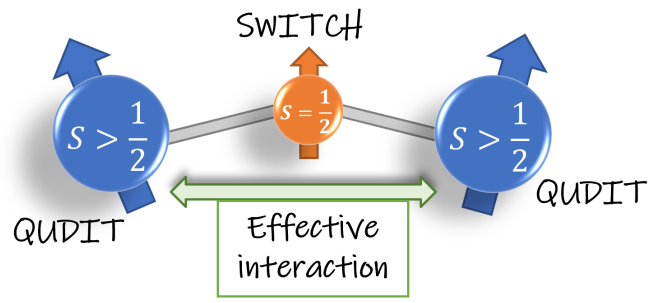

FIG. 1. Schematic representation of the elementary hardware structure of Q_1-S_w-Q_2. The central switch allows one to dynamically turn on and off an effective interaction between qudits.

is modeled as

$$H_{\mathrm{tot}} = \sum_{\substack{i=Q_1,\\ Q_2,Sw}} H_i + H_{Q_1,Sw} + H_{Q_2,S_w}, \tag{8}$$

where the first terms are the Hamiltonians of the three separated objects and $H_{Q_i,S_w} = J\mathbf{S}_i \cdot \mathbf{S}_{Sw}$ represents the spin-spin interaction (assumed isotropic) between the qudits and the switch. It is important to emphasise that having a qubit mediating the interaction between qudits provides an essential advantage over having two permanently coupled qudits, namely the ability to dynamically activate and deactivate the effective Q_1-Q_2 interaction. A description of the operations that we can implement is given in Sec. III A. In order to have a properly working switch a certain hierarchy of parameters must be present. In particular, the couplings of qudits with the switch must be much smaller than differences between the respective energy gaps, in order to ensure that the eigenstates of the global system are almost factorized. This regime guarantees that qudits can be independently controlled. The system eigenstates can be labeled by the expectation value of the local S_z operators $|m_{Q_1}, m_{Sw}, m_{Q_2}\rangle$, with $m_{Q_i} = -S, \ldots, S$ and $m_{Sw} = \uparrow, \downarrow$. On the other hand, a larger coupling would permit one to use shorter pulses, resulting in faster two-qudit gates. Thus, a good compromise between the two conditions must be found.

In the numerical simulations reported in Sec. IV, we consider the following parameters: the static magnetic field along the z axis is set to $B_0 = 1.3$ T. The two qudits consist of $S = 3/2$ ions with g values of 1.95 and 2.05 and zero-field splittings of $D = -2.4 \times 10^{-2}$ meV and -3.0×10^{-2} meV [64,65]. Although the precise values of parameters are not important, choosing slightly different values for the qudits makes them nonequivalent, ensuring that all transitions are distinct. A similar nonequivalence can be obtained by assuming the easy axes of the two qudits [here z, set by the last term in Eq. (7)] to be oriented differently from each other.

To ensure factorization between qudit and switch eigenstates in the idle phase, we consider as S_w a spin $s = 1/2$ with a large g value of 4, coupled to the qudits by $J = 0.8\ (1.0) \times 10^{-2}$ meV. Given the large g value, S_w must be regarded as an effective spin. For example, one could use two of the three eigenstates of a spin $S = 1$ with a large zero-field splitting,

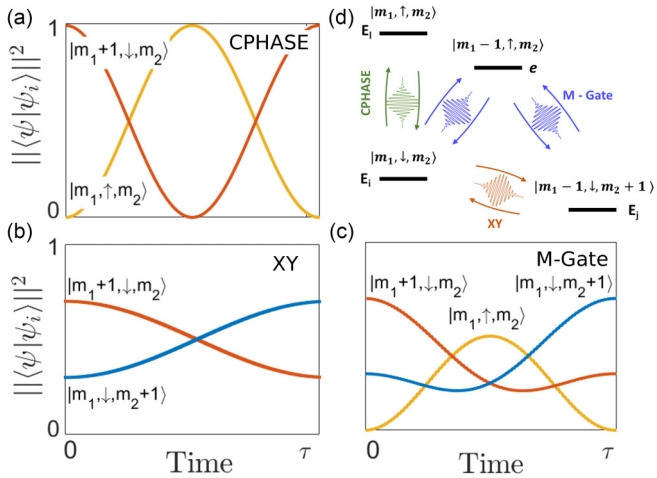

FIG. 2. Panels (a), (b), and (c) report the probability amplitudes of the different state components during the implementation of CPHASE(π), XY(2π), and M(2π) gates, respectively. In the case of the CPHASE gate, we initialized the system in the $|\psi\rangle = |m_1, \downarrow, m_2\rangle$ state, while for the other two gates we initialized in the $|\psi\rangle = 0.7|m_1, \downarrow, m_2\rangle + 0.3|m_1 - 1, \downarrow, m_2 + 1\rangle$ state. Panel (d) shows the connectivity scheme for the three introduced two-qudit gates.

such as $|m = 0\rangle$ and $|m = 1\rangle$ of a Ni^{2+} ion in a distorted octahedral environment.

A. Two-qudit gates

To implement a general algorithm, a universal quantum gate set is required. A minimal set can be obtained by combining single-qudit operations and an entangling two-qudit gate, such as a generalized control-phase gate [Eq. (6)], as shown in [5,61]. However, using a wider set of operations could lead to a shallower circuit decomposition and hence improve the hardware performance. In this work, we focus on gates that decompose and thus implement the time evolution of the operators necessary for the quantum simulations of fermionic systems, i.e., $\mathcal{H}_t$ and $\mathcal{H}_U$ in Eq. (1). In the following subsections we present three different two-qudit gates which can be used to decompose the time evolution.

1. Controlled-phase gate

The first gate we present is the generalized control phase gate (CPHASE), defined in Eq. (6): In our setup, a different phase $\varphi_{i,j}$ can be added to any level of the computational space by exciting end deexciting the state of the switch through a semiresonant pulse perpendicular to the direction of the static magnetic field [47,48,62,66] [vertical green pulse among $|m_1, \downarrow, m_2\rangle \longleftrightarrow |m_1, \uparrow, m_2\rangle$ levels in Fig. 2(d)]. In principle, if all the gaps are well resolved, this operation can be applied in parallel to all computational states. In this way, we are able to implement the most general CPHASE($\bar{\varphi}$) gate where a different phase is applied to all levels. As an example, in Fig. 2(a) we report probability amplitude evolution of the different state components $|\psi_i\rangle$ when adding phase π on the level $|m_1, \downarrow, m_2\rangle$. As can be seen, the population is transferred to the auxiliary level, initially not populated, and then brought back. In this way, the state acquires a specific phase. The pulse duration τ depends on the angle φ and can be explicitly determined as follows:

$$\Delta = \mathrm{sgn}(\varphi)\frac{2\Omega(\pi - |\varphi|)}{\sqrt{|\varphi|(2\pi - |\varphi|)}},$$
$$\tau = \frac{\pi}{\sqrt{\Omega^2 + \frac{\Delta^2}{4}}}, \tag{9}$$

where Ω is the Rabi frequency of the addressed transition, sgn is the sign function, and Δ is the frequency detuning. The gate duration has a strong dependence on the rotation angle and in particular it is maximum for $\theta = \pi$ and goes to zero for angles close to 0. For the system parameters considered in this work, and assuming a fast driving oscillating field $B_1 = 50$ G, a $\theta = \pi$ gate takes around 15 ns [see Fig. 4(a)].

2. XY gate

A single electromagnetic pulse parallel to the direction of the static magnetic field B_0 can induce a XY_{ij}^{lk} gate, Eq. 5, between any pairs of levels $|i, j\rangle$ and $|l, k\rangle$ that have the same total spin S_z^{tot} expectation value and differences between local spin operators ± 1 [47]. In other words, any pair that satisfies the relation $|i, j\rangle = |m_1, \downarrow, m_2\rangle \longleftrightarrow |l, k\rangle = |m_1 - 1, \downarrow, m_2 + 1\rangle$ is suitable [47] (see Fig. 2, panel d).

The mechanism by which this operation is generated can be interpreted by describing the global Hamiltonian as consisting of a leading term $\mathcal{H}_0$, which includes all the single-body terms and only the J^z part of the qudits-switch couplings, and by a perturbation term $\mathcal{H}_1$, which includes the transversal $J^{x,y}$ components of the couplings. The eigenstates of $\mathcal{H}_0$ are perfectly factorized. Moving to the rotating frame, the effect of an external driving field along z-axis resonant with the energy gap $E_{m_1,m_2} - E_{m_1',m_2'}$, is to compensate this gap, making the two states energetically degenerate [48,62,67]. Finally, the transverse coupling, which mixes the eigenstates, induces the transition. Alternatively, this mechanism can be described in the total Hamiltonian basis $\mathcal{H}_0 + \mathcal{H}_1$ on which the eigenstates are almost but not perfectly factorized on the S_z basis. Hence, the eigenstate identified by $|m_1, \downarrow, m_2\rangle$ actually contains a small component of even $|m_1 - 1, \downarrow, m_2 + 1\rangle$ and vice versa. So these eigenstates are connected by S_z and applying the driving field induces the transition.

In Fig. 2(b) we report the population evolution during the implementation of XY(π) between the levels $|m_1, \downarrow, m_2\rangle \longleftrightarrow |m_1 - 1, \downarrow, m_2 + 1\rangle$. The gate duration $\tau \propto \frac{\theta(E_i - E_j)}{J^2 B_1}$. Since the coupling constant J has to be small for the states to be almost factorised, the gate is usually very slow. Considering the parameters used in this paper ($B_1 = 50$G, $J \sim 10^{-2}$ meV), a complete exchange of populations (angle π) would take up to $\sim$1500 ns [see Fig. 4(a)]. Since the duration is directly proportional to the angle, this method is particularly suitable for small rotations.

3. M gate

We finally present a new procedure to implement an M_{ij}^{lk} gate, defined as

$$\begin{aligned}\mathrm{M}_{ij}^{lk}(\theta) = &\cos(\theta/2)(|i,j\rangle\langle i,j| - |l,k\rangle\langle l,k|)\\ &+ i\sin(\theta/2)(-|l,k\rangle\langle i,j| + |i,j\rangle\langle l,k|)\\ &+ \sum_{|r,t\rangle\neq|i,j\rangle,|l,k\rangle} |r,t\rangle\langle r,t|. \end{aligned} \tag{10}$$

This is very similar to XY, apart from phases. Therefore, they can be interconverted via a CPHASE gate of angle π: $\mathrm{XY}_{ij}^{lk}(\theta) = \mathrm{CPHASE}(\pi)^{lk}\mathrm{M}_{ij}^{lk}(\theta)$.

The M gate can be implemented by exploiting the quantum Householder reflection decomposition [61,68]. This transformation enables one to decompose any unitary d-dimensional gate into a sequence of simpler operations between pairs of computational levels. The transformation requires the use of an auxiliary level, i.e., a level outside the computational space, which must be connected to all involved computational levels. This type of connectivity is commonly referred to as a pod system. In this case, since the target unitary matrix is 2×2, its decomposition in very simple and requires a single operation. As done with the XY gate, we implement the operation between $|m_1,\downarrow,m_2\rangle \longleftrightarrow |m_1-1,\downarrow,m_2+1\rangle$ computational states. As a third auxiliary state we consider $|m_1-1,\uparrow,m_2\rangle$, which is outside the computational space. By means of electromagnetic pulses along the z axis, we are able to induce transitions between the two computational levels and the auxiliary one, and in this way form the desired pod system [see Fig. 2(d)]. Exploiting two simultaneous pulses and opportunely calibrating their relative pulse amplitudes and phases, the operation is implemented. In Fig. 2(c) we report the population evolution during the implementation of gate $\mathrm{M}(\pi)$ between the levels $|m_1,\downarrow,m_2\rangle \longleftrightarrow |m_1-1,\downarrow,m_2+1\rangle$, where the $|m_1-1,\uparrow,m_2\rangle$ state serves as auxiliary level. The gate duration τ is almost constant with respect to the rotation angle and takes $\sim$100 ns with the parameters considered in this work [see Fig. 4(a)].

In summary, we have identified three different two-qudit gates which can be implemented by magnetic pulses in the proposed molecular systems. As we will see below, the CPHASE gate turns out to be the fastest one [Fig. 4(a)]. XY and M gates are significantly slower, apart from very small angles where XY is competitive with CPHASE. Such small angles may occur when the time interval of a single Trotter steps is small.

IV. SIMULATIONS

A. 2D Hubbard-like model

To test and benchmark the proposed scheme, we implement a quantum simulation of four interacting spinless fermions on the elementary cell of a 2D lattice (see Fig. 3). The system is described by the following Hamiltonian:

$$\mathcal{H} = -\sum_{i,j=1,i\neq j}^{4} t_{i,j}a_i^\dagger a_j + \sum_{i=1}^{4} U_i a_i^\dagger a_i a_{i+1}^\dagger a_{i+1}, \tag{11}$$

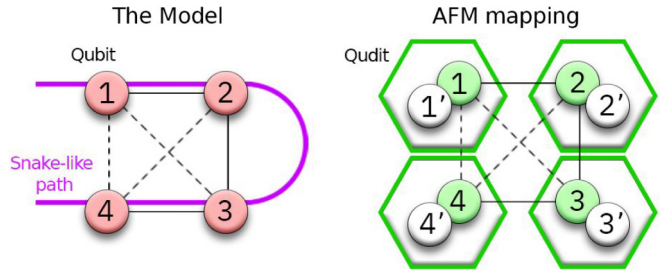

FIG. 3. Four-site fully connected spinless lattice. Left: the model where the black lines (dotted and continuous) represent hoppings between sites. The continuous ones are those along the linear labeling, while the dotted ones are those not along it. The continuous thick purple one highlights the index snakelike pattern. Right: AFM mapping where ancillary sites are introduced. The green hexagons enclose the qubits that are mapped into a single qudit.

where in the interaction term ($i=5\rightarrow i=1$). Compared to a standard Hubbard model we chose to remove the spin. This lowers the computational cost of our simulations without qualitatively changing the nature of the problem and the issues associated with the fermionic statistics. The interaction term describes a repulsion ($U_i>0$) between particles on nearest-neighbor sites. From the mapping and gate decomposition perspective, this term is analogous to the usual Fermi-Hubbard onsite interaction. The AFM mapping on qubits for this type of models is reported in Appendix B. To move to qudit mapping, as explained in Sec. II B, we map the product space of each site and relative auxiliary ones into a single qudit. To implement the quantum simulation, the Hamiltonian evolution terms have to be decomposed into elementary gates. The simplest terms are the interaction ones, which can be implemented directly by generalized control-phase gates. To understand this, it is easier to start by reasoning on the qubit basis. Considering a two-body interaction $e^{-i\mathcal{H}^{pq}\tau} = e^{-iU(a_p^\dagger a_p a_q^\dagger a_q)\tau}$, where p and q are the site indices, the action of the operator is to add a phase $\theta = U\tau$ to the states where orbitals p and q are both occupied. For the qubit AFM method, the interaction terms can be expressed in the form $A_p^\dagger A_{p\,\mathrm{tot}} I_{p'} A_q^\dagger A_q I_{q'}$ where the identity operators I act on auxiliary qubits. Hence, independently of the state of the two auxiliary qubits, a phase θ has to be added to the four states where p and q qubits are in the $|00\rangle$ state. It is not difficult to see that this operation, in terms of the proposed qudit mapping, results in four (parallelizable) generalized control-phase gates.

For the hopping terms, one can proceed in a similar manner. A hopping term $a_p^\dagger a_q + a_p a_q^\dagger$ is naturally expressed as a XY gate. In the simplest case, the hopping operators required by the mapping in the qubit basis are of the form $t(A_p^\dagger Z_{p'} A_q I_{q'} + A_p Z_{p'} A_q^\dagger I_{q'})$. These, in the qudit space, translate into four parallelizable generalized XY gates, two of them with an angle $\theta = t\tau$ and the others with an angle $-\theta$. The forms of other operators such as $t(A_p^\dagger X_{p'} A_q Y_{q'} + A_p X_{p'} A_q^\dagger Y_{q'})$ can be similarly deduced. The XY gate can also be rewritten in terms of the others two-qudit gates. Similarly to the case of qubits, a generalized XY gate can be decomposed into two generalized CPHASE gates and a few single qudit gates.

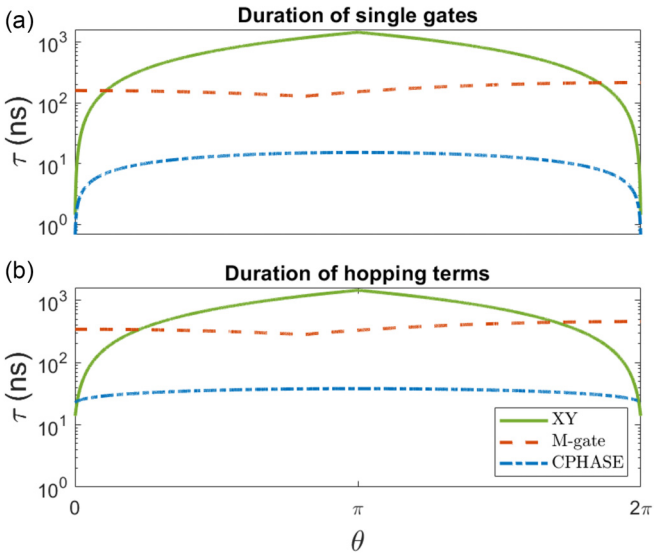

FIG. 4. (a) Comparison of gate duration as a function of angle (θ) for XY (continuous green line), M (dashed red line), and CPHASE (dotted blue line) gates. A driving field of 50 G is assumed. While the precise duration is dependent on the physical system details, the observed behavior and order of magnitude are generally valid. (b) Comparison of the duration for hopping terms as a function of rotation angle (θ) using CPHASE (blue), XY (green), and M (red) gates. A driving field of 50 G is assumed. This evaluation also takes into account single qudit gates.

Alternatively, an XY gate can be obtained by combining the M gate with a CPHASE to correct the phases.

The feasibility of this approach is demonstrated in the following sections by numerical simulations of molecular qudits. The effect of pure dephasing, which is the main error affecting MNMs at low temperatures, is taken into account considering a Lindblad dynamics (see Sec. IV B 2). In all the simulations and gate characterizations reported, we assume ideal pulses neglecting any effect of leakage. This choice is motivated by the fact that leakage is highly system specific and, if present, can be kept under control through pulse engineering [25,69–76] or quantum control techniques [77].

B. Results

We start by looking at single gates and hopping terms and then move to the whole system simulations. All the reported results are generated considering the hardware structure and parameters introduced in Sec. III.

1. Gate duration

First, it is interesting to explicitly compare the duration of the different operations we introduced in Sec. III A. This is of course a crucial aspect, as decoherence effects (included in the simulations of Sec. IV B 2) increase with the duration of gates. Figure 4(a) shows the duration of the different operations XY (continuous green line), M (dashed red line), and CPHASE (dotted blue line), as a function of the angle θ. First, we notice that for the considered system parameters the fastest operation for almost all angles is the CPHASE gate, while the relative performances of M and XY gates depend on the angle range. In particular, for small angles the duration of XY tends to zero, while it is maximum around π. Conversely, the M gate performs more or less the same for all angles and is therefore more efficient than XY for large angles and less efficient for small angles (considering angles wrap to π).

As mentioned above, by combining these operations with few single-qudit gates, it is possible to generate the time evolution associated with hopping and interaction terms. The interaction terms can be decomposed into a single CPHASE gate plus a few single qudit gates. Since a single-qudit gate is much faster than a two-body gate, the duration of the whole procedure is almost the same of a single CPHASE. For the hopping terms we have considered three possible decomposition strategies based on the three two-qubit gates introduced. A first possibility is to use two CPHASE combined with a few single qudit gates. This decomposition is analogous to the decomposition of the hopping terms on qubits and yields the fastest operation. Alternatively, we can decompose the hopping by XY or M gates. Since these two gates are not directly implementable between all pairs of levels, in some cases it is necessary to introduce several single-qudit π pulses before and after the two-body gate to temporarily exchange level populations.

In Fig. 4(b) we compare the performance of these gates, combined with single-qudit gates, in generating a hopping evolution. Overall, for most of the rotations angles, the fastest strategy is to use two CPHASE gates combined with few single qudit gates. The XY gate is competitive with CPHASE only for very small rotation angles, which may occur using a large number of Trotter steps. Although the CPHASE gate is the optimal choice for a two-qudit gate in the qudit Hamiltonian assumed here [Eq. (8)], it is still worthwhile to examine the extent to which slower gates affect the quantum simulation results. Thus, in the following we will perform simulations using the two other gates as well.

2. Decoherence modeling and gate characterization

In order to assess the actual feasibility of gates and simulations on real hardware, it is crucial to estimate the effect of decoherence. At low temperature, i.e., in the working conditions of a hypothetical hardware, the main effect of decoherence on the states of MNM qubits would be pure dephasing [64,65,78]. This phenomenon arises from the hyperfine interaction of the system spins with the surrounding nuclear spins, resulting in a decay of the off-diagonal elements of the density matrix. In contrast, phonon-mediated relaxation processes are negligible, since the associated relaxation time T_1 is several orders of magnitude larger than the dephasing time T_2. Indeed, in this temperature regime, phonon absorption is not allowed and the probability of phonon emission (scaling as the cube of the energy gap) is also irrelevant. In fact, we are interested in a set of states very close in energy to one another, with gaps considerably smaller than the Debye energy scale ($\sim$50 K).

The most accurate description of bath dynamics would take into account all microscopic interactions and would require a computationally expensive many-body method. However, a qualitative description that captures the essential behavior and greatly reduces the computational burden is possible. A common approach is to integrate a Lindblad master equation for the system as it evolves under the influence of the pulse sequence and decoherence. However, this method can

also be computationally expensive, even for relatively small systems. To further reduce the computational effort, we opted for an approximated approach that relies on quantum process tomography [51] (QPT) to characterize the elementary steps of the computation, which are then combined to build the whole algorithm.

QPT is a mathematical tool that allows one to create, in the superoperator formalism, a linear map $\mathcal{L}$ representing the action of a generic transformation on a quantum state. This map fully characterizes the process and allows one to obtain the output state of the register simply by applying the matrix $\mathcal{L}$ to the initial *vectorized* density matrix. In our case, we map the single- and two-qudit gates necessary for the quantum simulation, and we obtain the evolution due to the whole circuit by composing these elementary maps.

To obtain the map $\mathcal{L}$, we need to perform the quantum state tomography on a basis set for the matrix space. The set can be defined by choosing d^2 pure quantum states $\{|\Psi_1^0\rangle, |\Psi_2^0\rangle, \ldots, |\Psi_{d^2}^0\rangle\}$, where d is the dimension of the Hilbert space where the gate operates. These states are chosen such that the associated density matrices $\{|\Psi_1^0\rangle\langle\Psi_1^0|, |\Psi_2^0\rangle\langle\Psi_2^0|, \ldots, |\Psi_{d^2}^0\rangle\langle\Psi_{d^2}^0|\}$ form a basis set for the matrix space. Then, we numerically solve the Lindblad master equation for each density matrix $|\Psi_i\rangle\langle\Psi_i|^0$ of the basis and obtain the evolved matrix $|\Psi_i\rangle\langle\Psi_i|^\tau$ under application of the pulses needed for each gate. At last, we extract the superoperator $\mathcal{L}$,

$$\left|\Psi_i^\tau\right\rangle = \mathcal{L}\left|\Psi_i^0\right\rangle, \tag{12}$$

by solving the linear system.

QPT is a computationally expensive tool (requiring d^4 independent calculations), but, by characterizing all gates by their $\mathcal{L}$ map, it makes it easier to simulate the time evolution of the qudits under the effect of a noisy quantum circuit.

Having in mind the Q_1-S_w-Q_2 elementary unit described above in Eq. (8), in the secular and Born-Markov approximations the time evolution of the density matrix ρ is described by the equation

$$\begin{aligned}\frac{d\rho(t)}{dt} &= -\frac{i}{\hbar}[H_{\text{tot}} + H_1(t), \rho(t)] \\ &+ \frac{1}{T_2^{Q_1}}\left[2S_z^{Q_1}\rho(t)S_z^{Q_1} - \left(S_z^{Q_1}\right)^2\rho(t) - \rho(t)\left(S_z^{Q_1}\right)^2\right] \\ &+ \frac{1}{T_2^{S_w}}\left[2S_z^{S_w}\rho(t)S_z^{S_w} - \left(S_z^{S_w}\right)^2\rho(t) - \rho(t)\left(S_z^{S_w}\right)^2\right] \\ &+ \frac{1}{T_2^{Q_2}}\left[2S_z^{Q_2}\rho(t)S_z^{Q_2} - \left(S_z^{Q_2}\right)^2\rho(t) - \rho(t)\left(S_z^{Q_2}\right)^2\right],\end{aligned} \tag{13}$$

where H_{tot} is the free system Hamiltonian, $H_1(t)$ is the time dependent Hamiltonian describing the coupling with control electromagnetic pulses, and the last three terms describe the pure dephasing mechanism for Q_1, S_w, and Q_2, respectively. The T_2 parameter represents the dephasing time for superpositions of two levels with $|\Delta m| = 1$. If $|\Delta m|$ is larger, the dephasing effect of the incoherent terms grows quadratically as Δm^2.

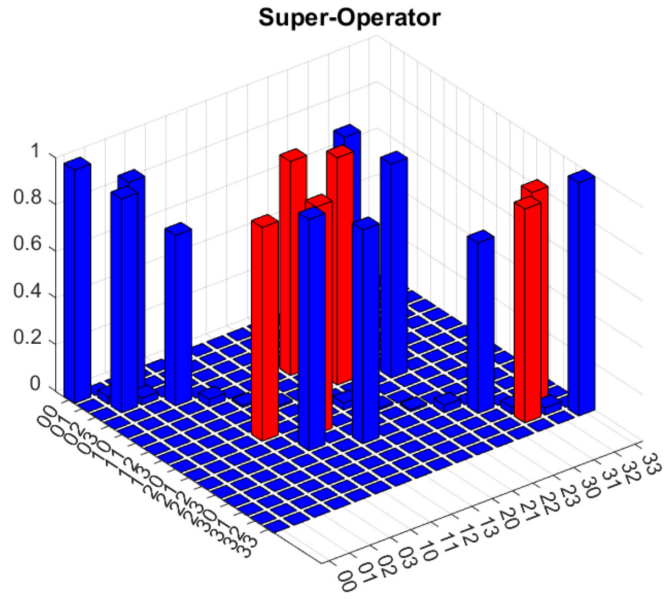

FIG. 5. Visual representation of $\mathcal{L}$ for a single qudit rotation between the second and third levels of a four-level qudit. Blue (dark gray) and red (light gray) bars identifies a relative phase of π. Simulations were performed considering $T_2 = 50$ ns and a driving field of 50 G.

For single-qudit gates, the equation is of course much simpler because only the involved qudit is considered.

As an example, we display in Fig. 5 the graphical representation of the $\mathcal{L}$ matrix for a single-qudit rotation that exchanges levels 1-2. Matrix entries specify the transformation applied to populations and coherences of a density matrix. In particular, the entries 00, 11, 22, 33 are for the populations of the states $|0\rangle$, $|1\rangle$, $|2\rangle$, $|3\rangle$, while all others are for coherences. For example, since the state $|0\rangle$ ($|3\rangle$) is not involved in the transformation, the 00 (33) goes into 00 (33). Differently, the population 11 (22) is transferred to 22 (11). Due to the effect of pure dephasing, the matrix presents some differences compared to the ideal one where the main bars should all be of height 1, and secondary small bars are absent. The bar color identifies a relative π phase.

It must be noted that this procedure assumes that the implementation of a gate does not influence the qudits that are not directly involved. This would require small variations of Hamiltonian parameters across the molecule-switch-molecule units that cannot be manipulated in parallel in the quantum simulation, in order to make these units spectrally distinguishable from one another. Yet, if differences in Hamiltonians are slight, the gates would operate in a very similar way on all units, resulting in tomographic maps very close to the ones we used.

3. Four-site model with qudit AFM

Finally, in this section we consider the simulation of the four-site spinless fermion model introduced above. We set $t = U$ for all hoppings on the square edges (1-2, 2-3, 3-4, 4-1) and $t = U/\sqrt{2}$ for those along the diagonals (1-3, 2-4) (other values would work equally well). The interaction term is parametrized by U and is only present between consecutive

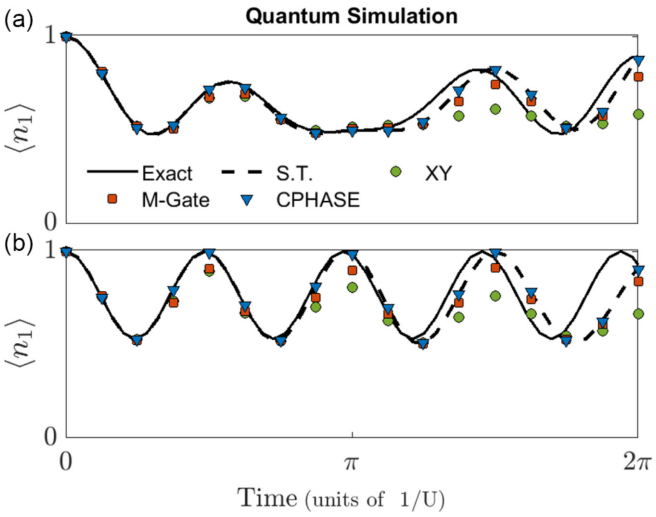

FIG. 6. Panel (a) [(b)] illustrates the evolution of the population on site 1, where sites 1-2 [1-3] were initially populated. The solid black line represents the exact solution, the dashed line depicts the Trotterized curve with $n_{\mathrm{TR}} = 6$, and the blue triangles, red squares, and green points correspond to simulations using CPHASE, M, and XY gates to implement hopping terms, respectively, considering a decoherence time $T_2 = 100$ µs.

sites. We consider two possible half-filled initial states: one with sites 1 and 2 occupied and the second with sites 1 and 3 occupied (top and bottom panels of Figs. 6 and 7, respectively). The simulation time is given in terms of the coupling parameter U, and all simulations are performed in the range 0–2π. For simplicity, we also consider a fully connected hardware coupling map, so no SWAP gates are introduced. Figure 6 shows the population evolution on site 1 for the two

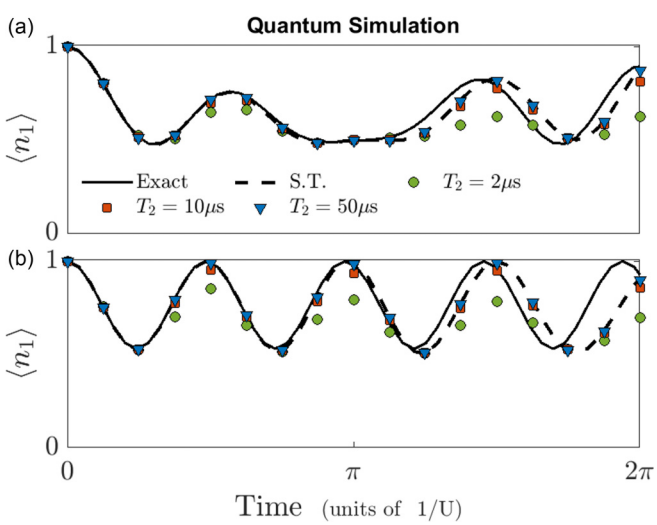

FIG. 7. Simulations of the four-site spinless interacting fermions illustrated in Fig. 3. The top (bottom) panel shows the evolution of the population at site 1, where sites 1-2 (1-3) are initially populated. The solid black line is the exact solution, the dashed line is the Trotterized curve with $n_{\mathrm{TR}} = 6$, and the green points, red squares, and blue triangles are simulated using CPHASE gates to implement hopping terms, and considering coherence times of $T_2 = 2, 10, 50$ µs, respectively.

initial states. The black continuous line is the exact solution, the dashed black line is the Trotter solution with $n_{\mathrm{TR}} = 6$ steps, and points are the simulation results by using CPHASE (blue triangles), M (red squares), and XY (green dots) gates to implement hopping terms. The top (bottom) panel reports the occupation probability on site 1, $\langle n_1 \rangle$, with the system initialized in a state where sites 1-2 (1-3) are populated. All simulations are performed assuming $T_2 = 100$ µs and a driving field of 50 G. While different two-qudit gates are used for the hopping terms, in all cases the interaction terms are decomposed by a single CPHASE gate. The whole circuit depth in terms of two-qudit gates is 30 using XY and 48 for M and CPHASE decompositions. In fact, each hopping term is decomposed into four parallel XY gates, resulting in a depth of 1, or two nonparallel generalized CPHASE gates plus some single qudit rotations, resulting in a depth of 2, or four parallel M gates plus a generalized CPHASE gate, giving a depth of 2. A single generalized CPHASE gate is used for each interaction term, giving a depth of 1. Since opposite couplings can be implemented simultaneously, three steps are required for hopping and two for interaction terms. Considering the six Trotter steps, one obtains the values given above. The qubit AFM decomposition of the same model would require a minimum depth of 72 CNOT gates even considering a fully connected coupling map.

In terms of physical implementation time, the circuits take about ∼500 ns for each Trotter step using the M gate. For CPHASE and XY gates, the implementation time depends on the simulation time and ranges from ∼150 to ∼300 ns for the former, and from ∼100 ns to ∼6 µs for the latter. Since the simulation time is up to 2π with six trotter steps, the largest angle required by hopping terms is $\pi/3$. This choice was made because the CPHASE and M gates had almost reached their maximum duration, while for the XY gate the effect of decoherence was already significant.

All the decomposition strategies effectively approximate the trotterized curve up to time $\sim \pi$. At longer times, however, a clear classification of the approaches emerges. Although CPHASE is not the most efficient in terms of number of gates, it emerges as the most effective strategy for implementing the simulation. The second-best approach is the M gate, while the XY gate lags behind. In particular, the XY gate struggles in the long-time region where large angles are required. However, the actual performance of these different strategies may vary depending on the architecture and the specific application. For example, in the case of an architecture with very long coherence times and a problem that requires high precision with short Trotter step, the XY gate approach may be advantageous. Therefore, the ability to implement different gate schemes is an interesting feature of the proposed setup.

Given the good results obtained with the CPHASE decomposition, Fig. 7 illustrates the simulation using only CPHASE and considering three shorter coherence times, $T_2 = \{50, 10, 2\}$ µs. With $T_2 = 50$ µs (blue triangles), the simulation results remain qualitatively robust, accurately reproducing the curve up to time π. However, with progressively shorter coherence times (red squares and green points) the effect of decoherence becomes more pronounced, leading to a decrease in the quality of the simulation results.

V. DISCUSSION AND CONCLUSIONS

Our study showcases the potential of a qudit extension of the auxiliary fermion method as an efficient strategy for simulating Hubbard-like Hamiltonians. A qudit-based encoding minimizes the number of two-body operations to be executed in series, thereby reducing circuit depth. This feature is a crucial factor for near-term quantum computing platforms.

Molecular nanomagnets are promising candidates for such applications thanks to their most prominent features, i.e., a multi-level structure and a high chemical tunability. Through realistic numerical simulations, we have demonstrated the feasibility of this approach using a supramolecular system as register. The diverse two-body control techniques explored in this work provide flexibility for adapting to various system architectures and target Hamiltonians. A natural extension of this work involves considering qudits encoded into the lowest energy states of multispin molecules with competing exchange interactions, inherently more resilient against decoherence compared to single-ion spin qudits, as recently demonstrated [26,61].

Full scalability and local control of a molecular qudits register will eventually require to extend this approach to a different paradigm, where single molecules or small supramolecular units are placed within superconducting resonators and photons are exploited for manipulation and readout [43,44,79]. This technology, already well developed for superconducting qubits, would allow architectures to be scaled and controlled much more easily than by using supramolecules. The weak coupling of the molecular spins with photons has been the greatest obstacle for the resonator approach so far, but a strong coupling regime appears to be within reach thanks to advances in resonator technology [43,44,80,81]. Although the setup proposed here is chemically difficult to scale up to many units, a small cluster could be used for a first proof-of-principle experiment in which an ensemble quantum simulation is performed.

ACKNOWLEDGMENTS

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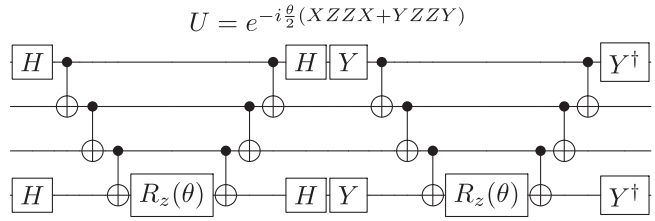
APPENDIX A: STANDARD TIME EVOLUTION OPERATORS DECOMPOSITION

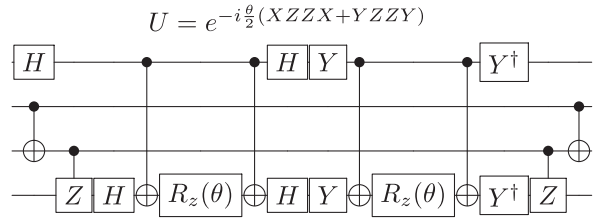
In this section, we briefly summarize the standard way to decompose the time evolution of hopping and interaction operators into one- and two-qubit gates [37]. By applying the Jordan-Wigner transform, a hopping interaction $a_p^\dagger a_q + a_q^\dagger a_p$ between sites p and q is rewritten as $A_p^\dagger(\bigotimes_{i=p+1}^{q-1} Z)A_q + A_p(\bigotimes_{i=p+1}^{q-1} Z)A_q^\dagger$, i.e., $X_p Z_{p+1} \cdots Z_{q-1} X_q + Y pZp + 1 \cdots Z_{q-1} Y_q$.

As an example, here we show the gate decomposition of the operator $e^{-i\frac{\theta}{2}(XZZX+YZZY)}$ through CNOT gates and a single qubit:

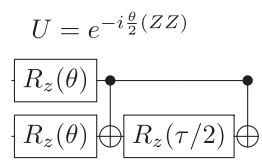


Here, H denotes the Hadamard gate, Y the Y-basis change and R_z a rotation around the z axis. This decomposition can be adapted to any Pauli sequence with minor adjustments. The main advantage of this method is that it requires only linear connectivity between qubits. However, a notable amount of not parallelizable CNOT gates $4(n-1)$ is required, where n is the length of the Pauli string.

By allowing CNOT between not consecutive qubits a more efficient decomposition is obtained. In this case the cost in terms of CNOT is $2n$:



Interaction terms such as $a_{p,\uparrow}^\dagger a_{p,\uparrow} a_{p,\downarrow}^\dagger a_{p,\downarrow}$ are mapped by JW into qubit operators as $A_{p,\downarrow}^\dagger A_{p,\downarrow} A_{p,\uparrow}^\dagger A_{p,\uparrow}$, which is explicitly rewritten as a ZZ term. The standard way to implement the time evolution of such operator is the following:



APPENDIX B: THE AUXILIARY FERMION METHOD

The auxiliary fermion method method was first proposed in 2005 by Vestrate and Cirac [40] and independently by Ball [41]. Later in 2016, Whitfield, Havlicek, and Troyer revisited it [42], clarifying some aspects and also giving an explicit practical formulation. Here, we provide an overview of the method and we examine the specific case studied in Sec. IV.

The idea behind the technique is to keep the parity information, relative to each coupling, stored in auxiliary qubits so that it does not have to be counted each time. In this way, the local fermionic operators can be mapped into limited size qubit operators. The additional cost in terms of qubits depends on the connectivity of the system. The benefit is a reduction in the length of the Pauli operators, which results in shallower circuits with fewer gates. The degree of the transformed operators, i.e., the number of qubits involved in an interaction, is related to the degree of the system graph and is independent of the lattice size. Therefore, for the same system graph complexity, AFM gives an increasing gain with increasing system size compared to JW or BK.

Before entering in technical details, we give a broad outline of the algorithm summarizing it into three main steps:

(1) Given a fermionic Hamiltonian and the relative graph representing sites and couplings, index sites along a snake-like path. For each site p, compute D_p as the number of non-local couplings, i.e. couplings not along the snake-like path, and introduce ceil$(D_p/2)$ additional auxiliary fermionic sites indexed adjacent to p.

(2) Add auxiliary stabilizer operators M_{aux}, acting only on auxiliary sites, to the original Hamiltonian and build an initial fermion state with desired fermionic statistics.

(3) Apply the Jordan-Wigner transformation to convert the fermionic Hamiltonian to qubit representation. Local fermionic operators are mapped to limited size qubit ones.

In the first step, the target Hamiltonian is represented as a graph, where each fermionic site corresponds to a vertex of the graph and each interaction is a link. A snakelike pattern is drawn over the graph numbering the vertices in ascending order. This is the same procedure used in JW mapping. When applying the JW mapping, Hamiltonian terms between two sites consecutive in the indexing are mapped to local qubit operators, whereas couplings between not-near indices are mapped to nonlocal qubit ones (Z chains). For each site p of the graph with at least one coupling not along the pattern, the AFM method requires introduction of auxiliary sites. Each site can be exploited to store the parity of at most two nonlocal links, therefore ceil$(D_p/2)$ additional auxiliary fermionic sites, where D_p is the number of sites coupled to p not along the linear indexing. For example, considering a Hubbard Hamiltonian on a 2D lattice with nearest-neighbor connections, a general site not on the edges would have four connections: two along the pattern and two not along it. Therefore, a single auxiliary site is required. Now, looking at the example case depicted in Fig. 3, where we have assumed to have hopping interactions among all sites, the linear pattern is highlighted with solid links (1-2, 2-3, and 3-4), while the other couplings are represented with dashed lines (1-3, 1-4, and $2-4$). All sites have one or two dashed links. Therefore, a single auxiliary orbital is introduced adjacent to the original one and marked by a $'$ symbol. The new linear indexing is $11'22'33'44'$.

In the second stage, the original Hamiltonian is modified by introducing stabilization operators M_{aux}, which act only on auxiliary sites, and are introduced to cancel the JW Z chains without changing the physics of the original fermionic model. For example, given a standard hopping term between sites p and q not adjacent on the indexing, the operator $a_p^\dagger a_q + a_q^\dagger a_p$ is replaced by $a_p^\dagger M_{\text{aux}}(pq) a_q + a_q^\dagger M_{\text{aux}}^\dagger(pq) a_p$. It is clear that by introducing auxiliary sites, and thus increasing the dimension of the Hilbert space, not every qubit state corresponds to a physical state in the fermionic system. To address this issue, it is imposed that the physical fermionic states on qubits correspond to $+1$ eigenstates of all the $M_{\text{aux}}(ij)$ operators. In this sense, these operators stabilize the physical subspace. Therefore, by construction, any pair of stabilizers commutes within this subspace. This implies that, for all p $[M_{\text{aux}}, a_p] = 0$, $a_p^\dagger M_{\text{aux}} a_q|\Omega\rangle = a_p^\dagger a_q M_{\text{aux}}|\Omega\rangle = a_p^\dagger a_q|\Omega\rangle$, where $|\Omega\rangle$ is the vacuum state. This condition enables the simulation of auxiliary fermions without affecting the original fermionic system. Note that in order to respect the fermionic statistics, the vacuum state $|\Omega\rangle$ must also be modified, as will be shown later.

The M_{aux} operators are defined to correspond to an observable with eigenvalues ± 1, so they are Hermitian and $M_{\text{aux}}^2 = \mathbf{1}$. These operators have the general form $M_{\text{aux}}(p,q) = i b_{p'} c_{q'}$ [see Eq. (4)] where θ and ϕ angles are set to respect the commutations rules mentioned above. Under the JW transformation, M_{aux} are rewritten as follows:

$$\begin{aligned} b_{p'} &= \left(\bigotimes_{i=0}^{p'-1} Z_i\right) \otimes B_{p'}, \\ c_{q'} &= \left(\bigotimes_{i=0}^{q'-1} Z_i\right) \otimes C_{q'}, \end{aligned} \tag{B1}$$

where $B_{p'} = [\cos(\theta)X + \sin(\theta)Y]_{p'}$ and $C_{q'} = [\cos(\phi)X + \sin(\phi)Y]_{q'}$. The Z chains introduced by the M_{aux} will cancel with the ones generated by the original Hamiltonian therefore resulting in local operators. Looking at the model above, the hopping part of the Hamiltonian is as follows:

$$\begin{aligned} H_t = -t\big(&a_1^\dagger a_2 + a_1 a_2^\dagger + a_2^\dagger a_3 + a_2 a_3^\dagger \\ &+ a_3^\dagger a_4 + a_3 a_4^\dagger + a_1^\dagger a_4 + a_1 a_4^\dagger \\ &+ a_2^\dagger a_4 + a_2 a_4^\dagger + a_1^\dagger a_3 + a_1 a_3^\dagger\big). \end{aligned} \tag{B2}$$

Directly applying the JW mapping, these would be transformed as

$$\begin{aligned} H_t = -t\big(&A_1^\dagger A_2 + A_1 A_2^\dagger + A_2^\dagger A_3 + A_2 A_3^\dagger \\ &+ A_3^\dagger A_4 + A_3 A_4^\dagger + A_1^\dagger Z_2 Z_3 A_4 + A_1 Z_2 Z_3 A_4^\dagger \\ &+ A_2^\dagger Z_3 A_4 + A_2 Z_3 A_4^\dagger + A_1^\dagger Z_2 A_3 + A_1 Z_2 A_3^\dagger\big). \end{aligned} \tag{B3}$$

As we can see, hoppings between sites not consecutive along the linear inducing involve Z chains. Due to the limited size of the toy model these are short, but they grow with the size of the system.

Conversely, applying the AFM method the fermionic Hamiltonian takes the form

$$\begin{aligned} H_t = -t(&a_1^\dagger a_2 + a_1 a_2^\dagger + a_2^\dagger a_3 + a_2 a_3^\dagger + a_3^\dagger a_4 + a_3 a_4^\dagger \\ &+ a_1^\dagger M_{\text{aux}}(14) a_4 + a_1 M_{\text{aux}}(14)^\dagger a_4^\dagger \\ &+ a_2^\dagger M_{\text{aux}}(24) a_4 + a_2 M_{\text{aux}}(24)^\dagger a_4^\dagger \\ &+ a_1^\dagger M_{\text{aux}}(13) a_3 + a_1 M_{\text{aux}}(13)^\dagger a_3^\dagger), \end{aligned} \tag{B4}$$

where a possible choice for the operators is $\theta_{13} = \phi_{13} = \theta_{24} = \phi_{24} = 0$ and $\theta_{13} = \phi_{13} = \pi/2$. Transforming the auxiliary operators into qubit operators by JW mapping, these becomes

$$\begin{aligned} M_{\text{aux}}(1,4) &= Y_{1'} Z_2 Z_{2'} Z_3 Z_{3'} Z_4 Y_{4'}, \\ M_{\text{aux}}(2,4) &= X_{2'} Z_3 Z_{3'} Z_4 X_{4'}, \\ M_{\text{aux}}(1,3) &= X_{1'} Z_2 Z_{2'} Z_3 X_{3'}. \end{aligned} \tag{B5}$$

The Hamiltonian can be explicitly rewritten in terms of qubits operators as follows:

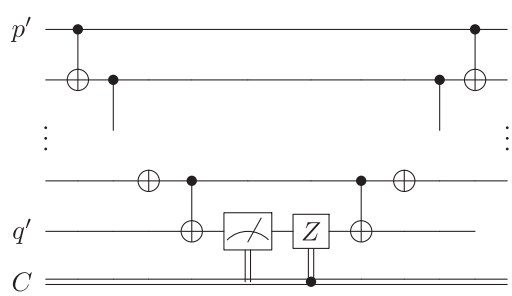
$$\begin{aligned} H_t = -t(&A_1^\dagger Z_{1'} A_2 + A_2^\dagger Z_{1'} A_1 + A_2^\dagger Z_{2'} A_3 + A_3^\dagger Z_{2'} A_2 \\ &+ A_3^\dagger Z_{3'} A_4 + A_4^\dagger Z_{3'} A_3 + iA_1^\dagger X_{1'} A_4 Y_{4'} - iA_1 X_{1'} A_4^\dagger Y_{4'} \\ &- iA_2^\dagger Y_{2'} A_4 X_{4'} + iA_2 Y_{2'} A_4^\dagger X_{4'} - iA_1^\dagger Y_{1'} A_3 X_{3'} \\ &+ iA_1 Y_{1'} A_3^\dagger X_{3'}). \end{aligned} \tag{B6}$$

The worst case locality, i.e., the largest number of sites involved in an interaction, is 4. We remark that this value is independent of system size, giving a huge advantage for large systems. For example, to simulate a Hubbard term of a square lattice with first-neighbor interactions of size $N = 30$ using JW mapping, the worst case locality would be $N + 1 = 31$, which in terms of circuit depth would require at least 62 CNOT, compared to 12 CNOT for the AFM. Furthermore, the overall circuit is much more parallelizable and can be implemented in just four steps.

Since the Hilbert space has been made larger, the vacuum state $|\Omega\rangle = |0, 0, \ldots, 0\rangle$ has to be rewritten as follows:

$$|\Omega\rangle = \prod_{(p,q)} \frac{1 + M_{\text{aux}}(p, q)}{\sqrt{2}} |0, 0, \ldots, 0\rangle. \tag{B7}$$

To build this state, we started from the state $|0, 0, \ldots, 0\rangle$ and then proceeded by projectively measuring each auxiliary coupling. If the outcome is -1 we apply a Z gate to p' or q' sites. In practice, this operation can be implemented in different ways with or without addition of ancillary qubits. A possible implementation that does not require additional qubits consists of storing the parity of the operator in a single qubit and then dynamically measuring it [see the scheme in (B8)]. If the outcome of the measurement is -1 a classically controlled Z gate is applied to restore the correct parity:



(B8)

The cost of this process scales like JW $\sim O(N)$. However, this price has to be paid only once at the beginning of the simulation and not at each step of it.

Once the vacuum state is created, we populate the orbitals by applying an X gate to the qubits. To keep the parity count correct, a Z gate is applied to any auxiliary site p' if the number of sites populated to its right along the linear indexing is odd.

APPENDIX C: SINGLE QUDIT GATES

In this paragraph, we focus on the decomposition of a unitary operation on a d-dimensional qudit into a sequence of elementary operations between pairs of levels, using the planar rotation (PR) decomposition method [13,82–84]. The elementary operation, from which the PR method takes the name, is the planar rotation $U_{\mu\nu}(\theta, \beta)$, also known as Givens rotations, defined as follows:

$$\begin{aligned} U_{\mu\nu}(\theta, \beta) = &\cos(\theta/2)(|\mu\rangle\langle\mu| + |\nu\rangle\langle\nu|) \\ &+ \sin(\theta/2)\big(|\nu\rangle\langle\mu| e^{i\beta} - |\mu\rangle\langle\nu| e^{-i\beta}\big) \\ &+ \sum_{\ell \neq \mu,\nu} |\ell\rangle\langle\ell|, \end{aligned} \tag{C1}$$

i.e., equal to the identity except for the elements in the intersection between the μth and νth rows and columns. Each PR [Eq. (C1)] between a pair of levels $|\mu\rangle$ and $|\nu\rangle$ connected by a magnetic-dipole matrix element ($\propto \sum_j g_j^\alpha s_j^\alpha$) is implemented by a transverse magnetic pulse resonant with the corresponding transition. A general matrix $W \in SU(d)$ can be decomposed in PRs as follows:

$$\begin{aligned} W = &\Lambda\big(e^{i\alpha_1}, e^{i\alpha_2}, \ldots, e^{i\alpha_{d-1}}, e^{-i\sum_{k=1}^{d-1}\alpha_k}\big) \\ &\times U_{1,2}(\theta_1, \beta_1) U_{1,3}(\theta_2, \beta_2) U_{2,3}(\theta_3, \beta_3) \cdots \\ &\times U_{1,d}\big(\theta_{\frac{(d-1)(d-2)}{2}+1}, \beta_{\frac{(d-1)(d-2)}{2}+1}\big) \\ &\times U_{2,d}\big(\theta_{\frac{(d-1)(d-2)}{2}+2}, \beta_{\frac{(d-1)(d-2)}{2}+2}\big) \cdots \\ &\times U_{d-1,d}\big(\theta_{\frac{d(d-1)}{2}}, \beta_{\frac{d(d-1)}{2}}\big), \end{aligned} \tag{C2}$$

where Λ is a diagonal matrix with the elements shown in brackets. The diagonal matrix Λ can be decomposed into a sequence of phase gates $P_{\mu\nu}(\alpha)$:

$$\begin{aligned} P_{\mu\nu}(\alpha) = &|\mu\rangle\langle\mu| e^{i\alpha} + |\nu\rangle\langle\nu| e^{-i\alpha} \\ &+ \sum_{\ell \neq \mu,\nu} |\ell\rangle\langle\ell|. \end{aligned} \tag{C3}$$

Each $P_{\mu\nu}(\alpha)$ can be implemented by a semiresonant pulse between pairs of eigenstates $|\mu\rangle$ and $|\nu\rangle$ [28].

This PR method can be straightforwardly applied to systems characterized by an all-to-all connectivity between the energy levels. Single-spin MNM qudits, such as the one considered in this work, allow operations only between consecutive energy levels. Therefore, if a direct transition between the μth and νth levels is not allowed, the corresponding $U_{\mu\nu}$ can be further decomposed into available transitions using additional π pulses [61]. In this way, the PR method can be adapted to any physical system, including those with a limited connectivity.

APPENDIX D: SUZUKI-TROTTER DECOMPOSITION

The time evolution of the total Hamiltonian is implemented trough the standard Lie-Trotter formula [85–88]. This method, also known as "Trotterization," allows us to decompose the global Hamiltonian evolution into local evolutions. For a given number of Trotter steps n the method introduces an error:

$$U(\tau) = e^{-i\sum_l \mathcal{H}_l \tau} = \left(\prod_l e^{-i\mathcal{H}_l \tau/n}\right)^n + O\left(\frac{\tau^2}{n}\right). \tag{D1}$$

Higher-degree formulas, known as "Suzuki-Trotter" [89], and other methods, known as "post-Suzuki," have also been

proposed [90,91]. In any case, the higher the accuracy requirement, the deeper and more complex the circuit representing the discretized evolution has to be.

The standard approach, and the one we used in the reported simulations, is to fix the number of Trotter steps n and characterize the corresponding gates and circuits for each time τ. The alternative approach is to set a fixed Trotter time step τ_1, implying a variable number n of steps at each time. By characterizing the circuit that represents the time evolution at τ_1, the time evolution at time $\tau_n = n\tau_1$ is then obtained by subsequently applying n times the circuit. This approach is more convenient for experimental applications as it requires the gates to be characterized only once.

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